INCH-POUND
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PERFORMANCE SPECIFICATION

CONNECTOR, OXYGEN MASK HOSE, NON-EJECTION TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.

This specification covers the non-ejection type mask hose connector assembly used in conjunction with the aviator oxygen mask or full-face fire fighter's mask.

2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, Wright-Patterson AFB OH 45433-7017 or e-mailed to Engineering.Standards@us.af.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 1660

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

BB-A-1034	Compressed Air, Breathing (For Guidance Only)

COMMERCIAL ITEM DESCRIPTIONS

A-A-59503	Nitrogen, Technical

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-27210	Oxygen, Aviator's Breathing, Liquid and Gas
MS22058	Connector, Oxygen Hose to Regulator

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS - AIR FORCE

56A3696	Gasket, Connector, Oxygen Mask to Regulator
57B3623	Ring, Retaining, Demand Mask to Regulator Tube, Internal

(Copies of these drawings are available online at https://af-jedmics.navair.navy.mil/, for users with approved access rights granted through the Joint Engineering Data Management Information and Control System (JEDMICS). Users not located on a *.mil network may contact the JEDMICS Help Desk via email at JEDMICS@us.af.mil for instructions on obtaining and completing the necessary forms.)

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B633	Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel (DoD adopted)

(Copies of this document are available online at https://www.astm.org/.)

SAE INTERNATIONAL

SAE-AMS-STD-595	Colors Used in Government Procurement
SAE-AMS-STD-595/37038	Miscellaneous, Flat, or Lusterless
SAE AMS2417	Plating, Nickel-Zinc Alloy (DoD adopted)

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(Copies of these documents are available online at https://www.sae.org/.)

2.4 Order of precedence.
Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification.
The connectors furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Recycled, recovered, environmentally preferable, or biobased materials.
Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life-cycle costs.

3.3 Materials.
All materials shall be suitably treated to resist corrosion due to electrolytic decomposition, fungus, salt spray, and any other atmospheric condition that may be encountered during operational use or storage. The use of toxic chemicals, hazardous substances, and ozone depleting chemicals (ODCs) shall be avoided, whenever feasible.

3.3.1 Nonmagnetic materials.
Except where magnetic materials offer definite advantages, nonmagnetic materials shall be used for all components.

3.3.2 Nonmetallic materials.
Any nonmetallic materials that are affected adversely by continued use with oxygen shall not be used. Plastics which crack, chip, or permanently set under a compressive load of 200 psi shall not be used.

3.3.3 Protective treatment.
Cadmium coating or plating shall not be used. Any parts requiring zinc plating shall be protected in accordance with SAE AMS2417 or ASTM B633 (see 6.4).

3.3.4 Wear resistant treatment.
All aluminum parts shall be treated to provide a wear resistant surface (see 6.4).

3.4 Design.
The oxygen mask connector shall consist of the main housing, one aircraft supply port with a restrictor valve that has disconnect anti-suffocation capability, and a hose connection with restraint pin as shown on Figure 1. The connector shall be complete when the parts are assembled with the main housing.

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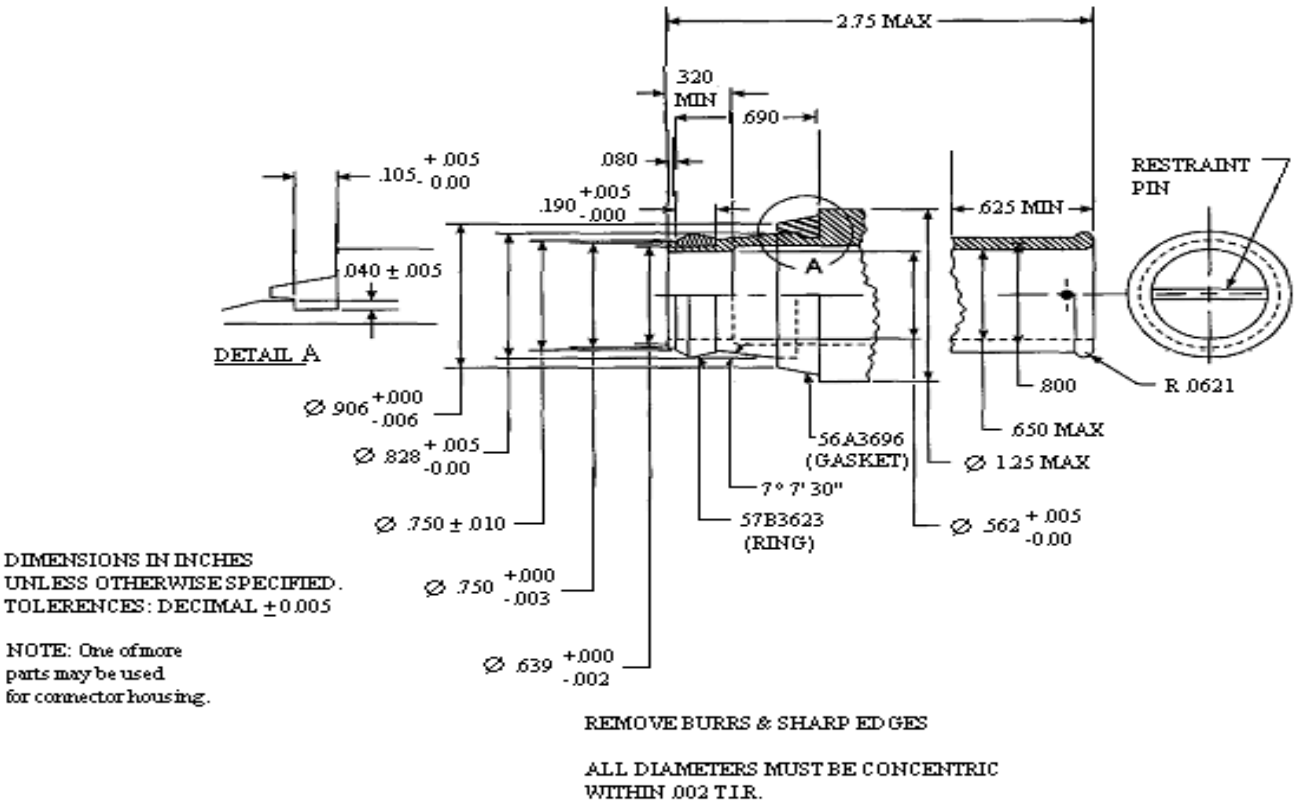


FIGURE 1. Mask port interface.

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3.5 Interface.

3.5.1 Aircraft supply port.
The aircraft supply port of the connector shall have a restrictor valve that has disconnect anti-suffocation capability, which permits the user to breathe normally when the restrictor valve is inserted into a connector conforming to MS22058. The restrictor valve shall be guided internally to prevent tilting or jamming of the valve mechanism. The valve actuator shall be positively locked to the restrictor valve.

3.5.2 Restraint pin.
A restraint pin shall be attached to the hose connection part of the connector to provide an attachment point for the oxygen mask hose restraint cord as shown on Figure 1.

3.5.3 Color.
The color of the connector shall be black, approximately matching color chip 37038 of SAE-AMS-STD-595.

3.6 Construction.
The connector shall be constructed so that no parts will become loose in service. It shall be built to withstand the strains, jars, vibrations, and any other conditions incident to shipment, storage, installation, and service.

3.6.1 Screw assemblies.
Assembly screws and bolts shall be tight. Tight shall be defined to mean that the screw or bolt cannot be appreciably tightened further without damage to the screw, bolt, or screw threads.

3.6.2 Installation of threaded parts.
All threaded parts shall be positively installed by self-locking nuts or any other approved methods so that the threaded parts will not work loose in service.

3.6.3 Lubrication.
Lubrication used in any component of the connector that maybe exposed to oxygen shall be approved by the procuring activity (see 6.5).

3.6.4 Cleanliness.
The connectors shall be free of oil, grease, debris, and any other contaminant that creates a hazard in an oxygen environment.

3.7 Performance.

3.7.1 Disconnection force.
The aircraft supply port of the connector shall disconnect from a jig conforming to MS22058 when a load of 12 to 20 pounds is applied along the longitudinal axis of the aircraft supply port.

3.7.2 Disconnection force reliability.
The disconnection force shall remain in the range specified in 3.7.1 for a minimum of 500 connection-disconnection cycles.

3.7.3 Leakage.
When subjected to an internal pressure of 1 psi, leakage of the connector shall not exceed 0.01 liter per minute (LPM) between -65 and 160 °F.

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3.7.4 Pressure drop through connector.
With the valve closed and pressure or suction applied to obtain a flow of 15 LPM from the regulator to the mask end, the pressure drop through the connector shall be within 4 to 6 inches of water.

3.7.5 Pressure drop through connector with jig.
With the connector mated to jig that has internal dimensions of the connector specified in MS22058, the pressure drop through the assembly shall not exceed the values shown in Table I.

TABLE I. Pressure drop in assembly.

Flow of Oxygen (LPM)	Pressure Drop (Inches of Water)
135	1.7
90	0.80
70	0.50
50	0.32
30	0.10
0	0

3.7.6 Reliability.
The connector shall have a minimum mean time between failures (MTBF) of 600 hours in order to provide a 16-hour mission reliability of 0.98.

3.7.7 Longevity.
The connector shall have a minimum operating life span of 2,000 hours.

3.7.8 Restraint pin.
The minimum force to separate the restraint pin from the connector shall be 110 pounds.

3.7.9 Odor.
The connector shall have no objectionable odors during operational use.

3.8 Item Identification.
The connector shall be permanently marked with the following information:

Connector, Oxygen Mask Hose
Manufacturer's Part Number
Contract Order Number
Manufacturer's Name or Federal Supply Code for Manufacturers (FSCM)

3.9 Interchangeability.
All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable.

4. VERIFICATION

4.1 Classification of inspections.
The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

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b. Conformance inspection (see 4.3).

4.2 Qualification inspection.
Qualification inspection shall consist of all tests specified in 4.6 except 4.6.2. The disconnection force test specified in 4.6.2 is a conformance inspection only.

4.2.1 Qualification samples.
The qualification samples shall consist of three connectors prepared in accordance with this specification and representative of production.

4.3 Conformance inspection.
Conformance inspection shall consist of:

a. Individual inspection (see 4.3.1).
b. Sampling inspection (see 4.3.2).

4.3.1 Individual inspection.
Each connector shall be subjected to the following inspections:

a. Examination of product (see 4.6.1).
b. Disconnection force (see 4.6.2).

4.3.2 Sampling inspection.

4.3.2.1 Sample size.
The sample size shall be three connectors chosen at random from each lot of 100 or less connectors.

4.3.2.2 Sampling tests.
The samples shall be subjected to the following tests:

a. Leakage, except that only the 77 ±18 °F portion of the test shall be conducted (see 4.6.4).
b. Pressure drop through connector (see 4.6.5).
c. Odor (see 4.6.9).

4.4 Test conditions.

4.4.1 Atmospheric conditions.
Unless otherwise specified in the acquisition documents (see 6.2), all tests required by this specification shall be made at an atmospheric pressure of 28 to 32 inches of mercury, at a temperature of 77 ±18 °F, and a relative humidity of 80 percent or less. When tests are conducted with atmospheric pressure or temperature substantially different from these values, proper allowance shall be made for the change in instrument reading.

4.4.2 Gas.
The gas used in testing the connectors shall be oxygen conforming to Type I of MIL-PRF-27210, water-pumped nitrogen conforming to A-A-59503, Type I, Grade B, Class 1, or compressed air conforming to BB-A-1034 (for guidance only), compressed source II, Grade B. The water content

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shall be 0.3 milligrams per liter at dew point -65 °F or 26.3 parts per million. If either nitrogen or air is used, appropriate density correction factors shall be applied to the flowmeter and to the performance of the connector.

4.5 Requirements cross-reference matrix.
Table II provides a cross-reference matrix of section 3 requirements tested or verified in section 4.

TABLE II. Requirements cross-reference matrix.

REQUIREMENT	VERIFICATION	REQUIREMENT	VERIFICATION
3.1	4.2	3.6.3	4.6.1
3.3	4.6.1	3.7.1	4.6.2
3.3.1	4.6.1	3.7.2	4.6.3
3.3.2	4.6.1	3.7.3	4.6.4
3.3.3	4.6.1	3.7.4	4.6.5
3.3.4	4.6.1	3.7.5	4.6.6
3.4	4.6.1	3.7.6	4.6.7
3.5.1	4.6.1	3.7.7	4.6.8
3.5.2	4.6.1	3.7.8	4.6.10
3.5.3	4.6.1	3.7.9	4.6.9
3.6	4.6.7, 4.6.8	3.8	4.6.1
3.6.1	4.6.1	3.9	4.6.1
3.6.2	4.6.1		

4.6 Tests.

4.6.1 Examination of product.
Each connector shall be examined to determine conformance with this specification with respect to materials, design, interface, lubrication, markings, interchangeability, and tightness or locking of threaded parts. There shall be no evidence of cuts, nicks, or holes in the connector hose.

4.6.2 Disconnection force.
A jig that has the internal dimensions of the connector specified in MS22058 shall be used for this test. The connector shall be inserted and disconnected from the aircraft supply port by applying sufficient load along the longitudinal axis of port. The load shall be measured to determine conformance with the load requirements of 3.7.1. This test shall be performed three times on each connector.

4.6.3 Disconnection force reliability.
The jig specified in 4.6.2 shall be used for this test. The connector shall be inserted and disconnected from the aircraft supply port at a rate of 20 cycles per 24-hour period for a total of 25 periods. The disconnection force for each cycle during the test shall meet the requirements of 3.7.1. The disconnection force for the three initial and three final cycles of each 20-cycle run shall be averaged and reported.

4.6.4 Leakage.
The jig that is described in 4.6.2 shall be used for this test. The hose connection port shall be blocked by using an elastomeric cap. An internal pressure of one psi shall be applied at temperatures of 77 ±18 °F, -65 ±2 °F, and 160 ±2 °F. The total leakage shall not exceed 0.01 LPM.

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4.6.5 Pressure drop through connector.
With the valve closed, pressure or suction shall be applied to the connector to obtain a flow of 15 LPM from the regulator end to the mask end. The pressure drop through the connector shall be 4 to 6 inches of water. The pressure tap shall be a suitable device (see 6.6) located on the mask side if suction is used and on the regulator side if pressure is used. The flow duct shall be constructed to ensure a straight flow for at least 6 inches immediately after leaving the connector if suction is used or for at least 6 inches immediately before entering the connector if pressure is used. The connector housing shall be mounted in a vertical position and its location shall permit free flow into the connector if suction is used and free flow out of the connector if pressure is used.

4.6.6 Pressure drop through connector with jig.
The connector shall be connected to the same jig as in 4.6.2. With the valve actuator located at the top of the slots, pressure shall be applied to the jig to obtain flows in the range of 0 to 135 LPM from the regulator end to the mask end. The pressure drop through the assembly shall be measured by a suitable device such as described in 6.6. Pressure drops through the assembly shall not exceed the values specified in Table I.

4.6.7 Reliability.
If required in the acquisition documents (see 6.2), the sample of connectors used for this test shall be subjected to simulated breathing cycles at a rate of 10-20 breathing cycles per minute at room temperature (77 ±18 °F). Cyclic breathing of the connectors shall be accomplished with a peak flow rate of 30 LPM inhalation and exhalation, with the restrictor valve in the open position. During 20 percent of the time that the connectors are subjected to breathing cycles, they shall be vibrated at a frequency ranging from 500 to 2,500 cycles per minute, at double amplitude of not less than 0.018 inch nor more than 0.020 inch. The connector shall be vibrated alternately in each of the three axes for 6 hours per axis. At least once every 24 hours of the breathing cycles, each connector shall be subjected to and shall meet the inspections specified in 4.6.2 (room temperature), 4.6.4, and one 20-cycle run as described in 4.6.7. No parts shall be replaced as preventative maintenance during reliability testing. Sufficient testing shall be conducted to demonstrate an MTBF of 600 hours at a confidence level of 0.9. At the conclusion of this testing the connectors shall be subjected to inspection as specified in 4.3.1.

4.6.8 Longevity.
If required by the acquisition documents (see 6.2), the operating life span shall be verified by extending the reliability test to reach 2,000 hours of operation by each connector.

4.6.9 Odor.
Gaseous oxygen shall be allowed to pass through the assembled connector at a rate not greater than 10 LPM for one minute. A smell test shall be conducted in a manner that shall prevent exterior odors from influencing the test. Using a minimum of six persons as testers, no more than 33 percent of the testers shall judge any odor as objectionable.

4.6.10 Restraint pin.
The restraint pin shall be pulled at its midpoint while installed in the hose connector until the pin breaks or deforms to such an extent as to become dislodged from its position. The force required to cause this failure or deformation shall not be less than 110 pounds.

5. PACKAGING

5.1 Packaging.
For acquisition purposes, the packaging requirements shall be as specified in the contract or order

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(see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

The connector covered by this specification is intended for oxygen masks in use on non-ejection aircraft and with full-face fire fighter's masks.

6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. If qualification is required (see 3.1).
c. If tests are to be conducted in other than atmospheric conditions (see 4.4.1).
d. If a reliability test is required (see 4.6.7).
e. If a longevity test is required (see 4.6.8).
f. Packaging requirements (see 5.1).

6.3 Qualification.

With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 83867 whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from 648 AESS/TAAM, Brooks City-Base, San Antonio TX 78235 (see footnote in QPD for QA info). An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil/.

6.3.1 Retention of qualification.

To retain qualification, the manufacturer must forward certification at two-year intervals to the qualifying activity stating that the company still has the capabilities and facilities necessary to produce the item and that the product has not changed in any way. The qualifying activity will establish the initial reporting date.

6.4 Protective and wear resistant treatment.

The following specifications may be used as a guide for 3.3.3 and 3.3.4:

MIL-DTL-5002	Surface Treatments and Inorganic Coatings for Metal Surfaces of Weapon Systems

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MIL-PRF-8625 Anodic Coatings for Aluminum and Aluminum Alloys (see Type III Coatings).

6.5 Lubrication.
MIL-PRF-27617, Grease, Aircraft and Instrument, Fuel and Oxidizer Resistant, is approved for use in oxygen systems.

6.6 Pressure measurement.
A piezometer ring located 1.5 to 2.5 inches from the mask end of the connector has historically been used to read pressure. For pressure drops through the connector including the assembly in a jig, a piezometer ring may be located approximately 2 inches from the end of the connector.

6.7 Subject term (key word) listing.

Aircraft supply port
Disconnection
Firefighter
Flange
Jig
Leakage
Mounting plate
Odor
Pressure
Supply port
Threads
Welding

6.8 Changes from previous issue.
Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

(Project 1660-2016-002)

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